

chemistry, among others.

Community contributions. As an open source project, Qiskit has a robust and active community, ensuring continuous improvement and support.

Google's Cirq

Cirq is an open source quantum software library developed by Google for designing and simulating quantum circuits on quantum processors, particularly NISQ (Noisy Intermediate-Scale Quantum) devices. Its key features are:

- Especially optimised for Google's quantum processors like Sycamore.
- Tools for constructing quantum algorithms and experimenting with them on simulators.
- Focus on quantum hardware and the challenges of noisy, intermediate-scale quantum devices.

Microsoft's Quantum Development Kit (QDK)

Microsoft's Quantum Development Kit includes the Q# programming language and tools to develop quantum algorithms. It also offers the Quantum Simulator for testing quantum algorithms without actual quantum hardware. Its main features are:

Q# programming language. This high-level language for quantum algorithms integrates with .NET languages like C#.

Quantum simulators. QDK includes simulators for small-scale circuits and resource estimation.

Quantum integration. Runs algorithms on quantum hardware via Azure Quantum.

Libraries for quantum algorithms. Has predefined libraries for quantum chemistry, machine learning, and cryptography.

Hybrid quantum-classical computing. QDK supports hybrid algorithms combining quantum and classical computations.

Cross-platform. Works on Windows, macOS, and Linux,

integrating with the .NET ecosystem.

Rigetti's Forest SDK

Forest is an open source quantum computing platform from Rigetti, offering a suite of tools for programming, simulating, and accessing quantum hardware via the cloud. Its key features are:

- Includes the Quil programming language and tools for developing quantum algorithms.
- Access to Rigetti's quantum computers through the cloud (via the Forest Cloud service).

MICROSOFT'S QUANTUM DEVELOPMENT KIT INCLUDES THE Q# PROGRAMMING LANGUAGE AND TOOLS TO DEVELOP QUANTUM ALGORITHMS. IT ALSO OFFERS THE QUANTUM SIMULATOR FOR TESTING QUANTUM ALGORITHMS WITHOUT ACTUAL QUANTUM HARDWARE.

- Simulators for testing quantum circuits before deploying them on real hardware.

PennyLane by Xanadu

PennyLane is an open source quantum machine learning (QML) framework developed by Xanadu, designed to combine quantum computing with machine learning and optimisation techniques. Its salient features are:

- Integration with popular machine learning libraries like TensorFlow and PyTorch.
- A focus on differentiable quantum programming for quantum machine learning tasks.
- Supports a wide range of quantum hardware backends.

How to get started with quantum computing using open source

Start with the basics. Begin by understanding quantum mechanics fundamentals, such as qubits, superposition, and entanglement. Familiarise yourself with quantum gates, circuits, and algorithms.

Work with simulators first.

Before accessing actual quantum hardware, use simulators available on platforms like Qiskit, Cirq, or Rigetti's Forest. These tools allow you to experiment without needing physical quantum machines.

Follow tutorials and examples.

Follow beginner tutorials and build simple quantum algorithms. Many open source frameworks, like Qiskit and Cirq, offer beginner-friendly examples to guide you through your first projects.

Join the community. Engage with online forums and communities (like Qiskit's community, Stack Exchange, and GitHub) where you can ask questions, share your progress, and collaborate with other learners.

Focus on small projects. Start with small quantum programming projects, such as creating a quantum circuit that solves simple problems like a quantum random number generator or implementing Grover's search algorithm. This will help you build your understanding progressively.

I encourage you to explore the exciting world of open source quantum computing. Whether you're a beginner or an expert, there are numerous projects, tutorials, and communities to help you get started. Contributing to these initiatives will not only advance your own understanding but also support the growth of a global quantum ecosystem. Start today—your participation could help shape the future of quantum computing! **EFY**

This article was first published in January 2025 issue of Open Source For You magazine.